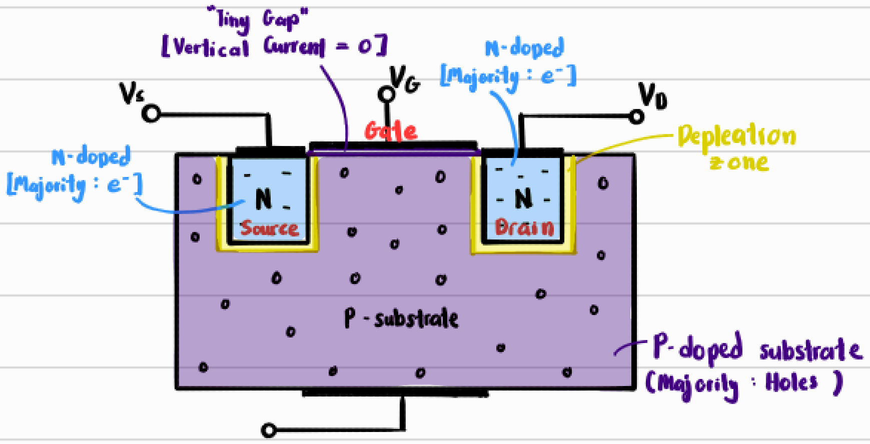
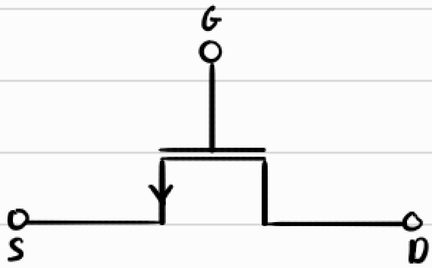
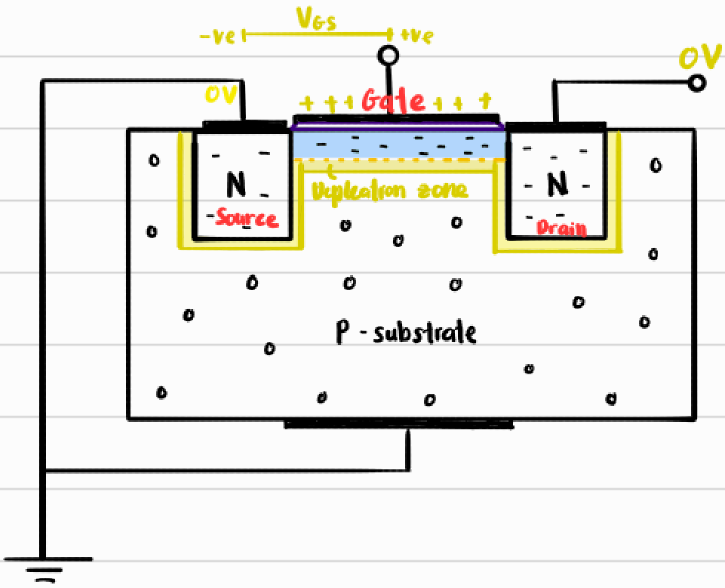
Metal Oxide Semiconductor Field Effect Transistor, MOSFET





① Channel creation



① As V_{GS} ↑, it pulls e^- to the gate, filling up the holes near the gate

② At a certain V_{GS}, all holes will be filled, this V_{GS} is called Voltage threshold, V_{th}.

③ As V_{GS} ↑ pass V_{th}, free moving e^- will accumulate near the gate creating a channel of free moving electrons. [N - channel]

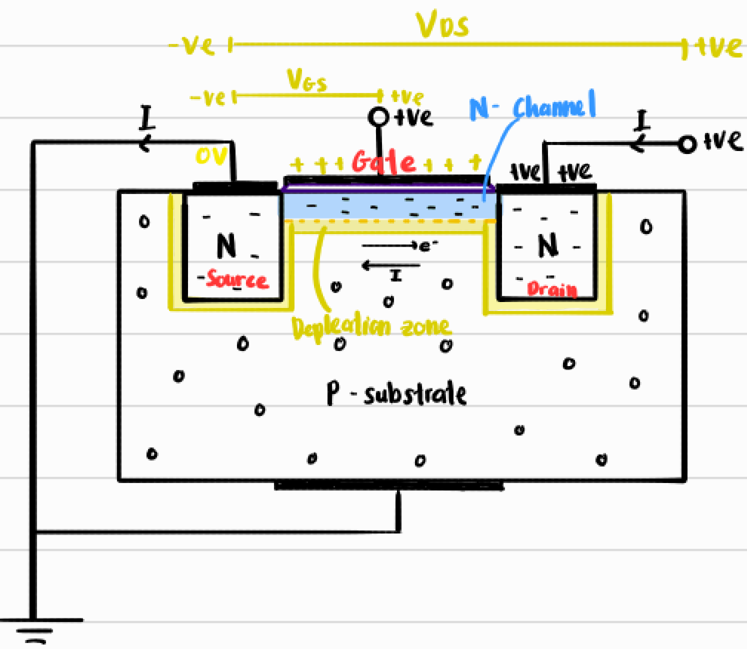
Now e^- can move from source to drain if a pd accross source & drain is applied

④ The depletion zone around source & drain vanished bcuz V_{GS} is high enuf to pull out e^- inside the holes

⑤ The depleation zone across the channel appeared bcuz the e^- near the p-substrate fill up the holes in the p-substrate

When $V_{GS} < V_{th}$, its in cutoff state

② Current flow



① When V_{DS} is applied, the moving e^- will flow to the drain [Source → Drain]

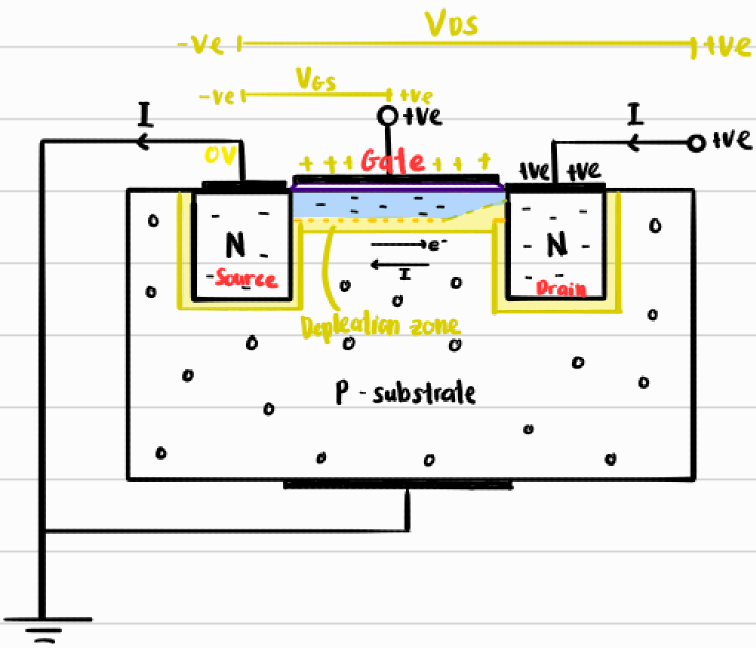
② Thus current will flow in the opposite direc. [Drain → Source]

③ When $V_{DS} < V_{GS} - V_{th}$, the MOSFET is in triode state

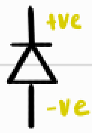
④ In triode state, as V_{DS} ↑, I ↑

Sometimes I is called drain current, I_D because the current comes from the drain

③ Pinch off effect



Notice that the N - channel and p-substrate acts like a diode facing up



① As $V_{DS} \uparrow \Rightarrow V_D \uparrow$, the diode is in reversed biased, causing the depletion zone near V_D to widen

② This constricts the channel, thus reducing the flow of e^-

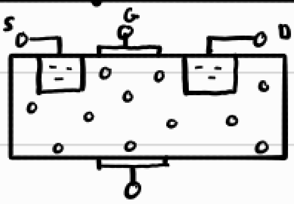
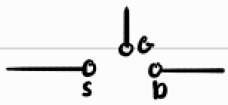
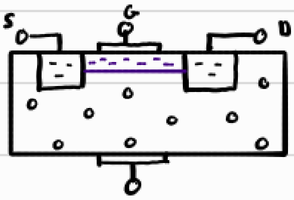
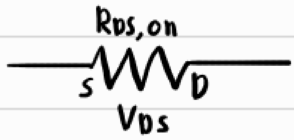
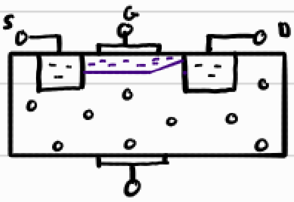
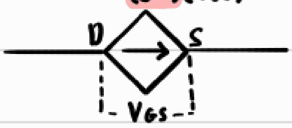
③ Due to the large flow of e^-, its impossible for the channel to be constrict ALL the e^-.

④ At a certain V_{DS}, the e^- will be saturated, and the flow of e^- will be constant & not reduce further w/ a further increase of V_{DS}.

↳ This is the saturation state

⑤ The only way to increase I, is to widen the channel by increasing V_{GS}. As $V_{GS} \uparrow$, $I \uparrow$

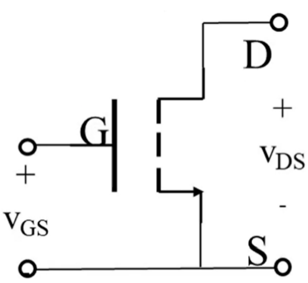
Simplified circuits

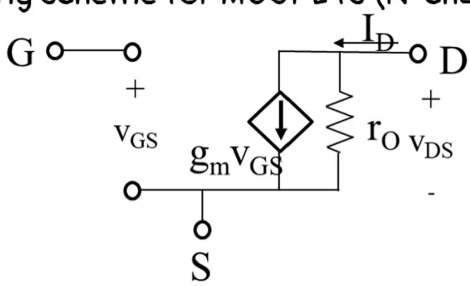
States	Original	Simplified
Cutoff [$V_{GS} < V_{th}$]		 Open circuit
Triode [$V_{GS} > V_{th}$] [$V_{DS} < V_{GS} - V_{th}$]		 Closed circuit
Saturation [$V_{GS} > V_{th}$] [$V_{DS} > V_{GS} - V_{th}$]		$I = (g_m)(V_{GS})$  Dependent current Dependent on V_{GS}

Transconductance parameter
$g_m = \frac{dI}{dV_{GS}}$
NIS

Signal analysis for MOSFET

Can use a similar modelling scheme for MOSFETs (N-channel for this example..)





$$r_O = \frac{1}{\lambda I_D}$$

Change from BJT – open circuit instead of rπ
For gm, need diD/dvGS (transconductance)
1/λ – from the channel length modulation effect

$$i_{D,sat} = \mu_n \frac{W}{L} . C''_{ox} \left(v_{GS} - V_{TN}\right)^2 \left(1 + \lambda . v_{DS}\right)$$

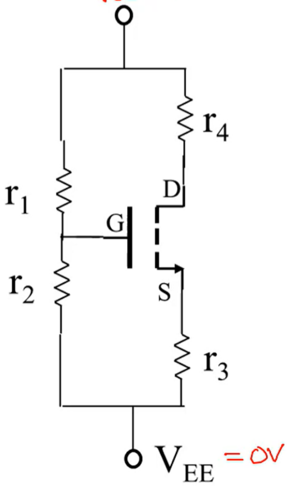
$$\Downarrow$$

$$g_m = \frac{di_D}{dv_{GS}} = \mu_n \frac{W}{L} C''_{ox}(V_{GS} - V_{TN})^2(1 + \lambda v_{DS}) = K_n(V_{GS} - V_{TN})^2(1 + \lambda v_{DS})$$

$$g_m = \frac{2I_D}{V_{GS} - V_{TN}}$$

Example 10.3

Write the KVLs for the four ~~transistor~~ resistor biasing of MOSFET shown in the diagram, where Vcc = 10V, R1 = 3kΩ, R2 = 2kΩ, R3 = 5kΩ, R4 = 20kΩ.



$V_{bias} = 10V$

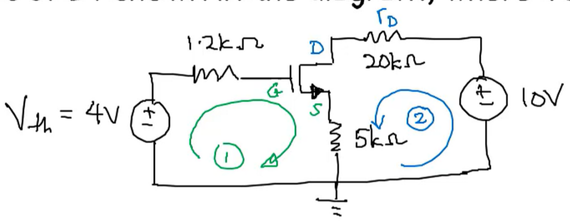
$V_{th} = \frac{2k}{2k + 3k} \times 10V$

$= 4V$

$R_{th} = r_1 \parallel r_2$

$= \frac{2k \times 3k}{2k + 3k}$

$= 1.2k\Omega$



KVL① : $+4 - V_{r_{th}} - V_{GS} - V_{r_S} = 0$

$4 - \cancel{I_G}^{0}(1.2k) - V_{GS} - I_S(5k) = 0$

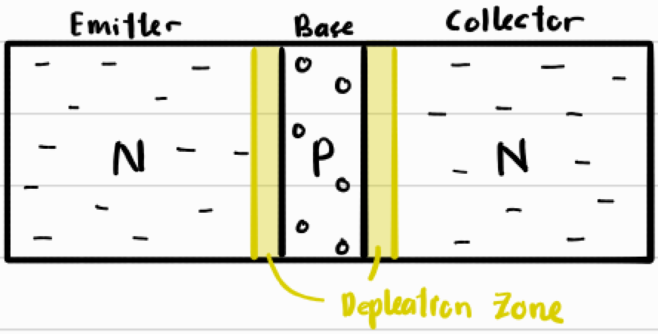
$4 - V_{GS} - I_S(5k) = 0$

$\rightarrow I_S = K_n(V_{GS} - V_{TN})^2(1 + \lambda V_{DS})$

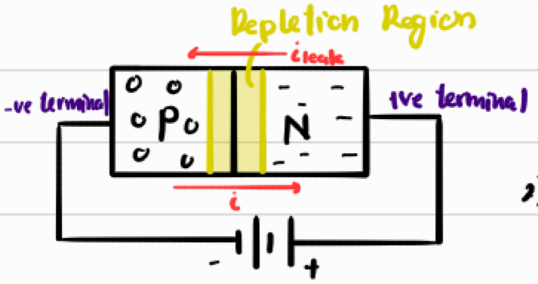
KVL② : $+10 - I_S(20k) - V_{DS} - I_S(5k) = 0$

V_{GS} ? V_{DS} ? $V_{DS} > (V_{GS} - V_{TN})$

Bipolar Junction Transistors [BJT]



First Recall how a Diode in Reverse Bias work

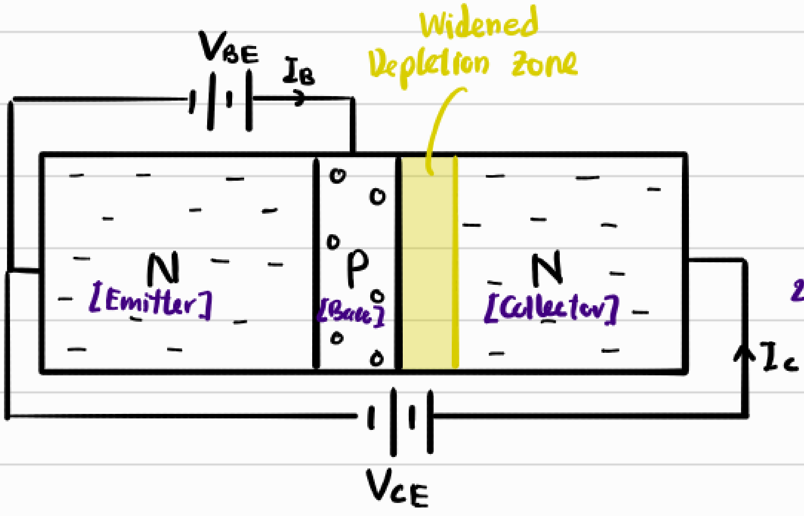


1) Initial Depletion Region caused by holes & e^- recombining at the boundary
2) Due to the reverse bias orientation,
   - e^- in N gets "pulled" towards +ve terminal
   - holes in P gets "pulled" towards -ve terminal

   This widens the depletion region.
3) e^- in N won't be able to jump to the holes in P
   $\Rightarrow i = 0$
4) However the minority charge carriers in P [e^-] will actually get sucked right across the depleation region into N
   $\Rightarrow i_{leak} \neq 0$

However, the quantity of e^- in P is very few, thus $I_{leak} \approx 0$.
What if we P actually had enough e^-?
That is what BJT Does.

BJT working mechanism



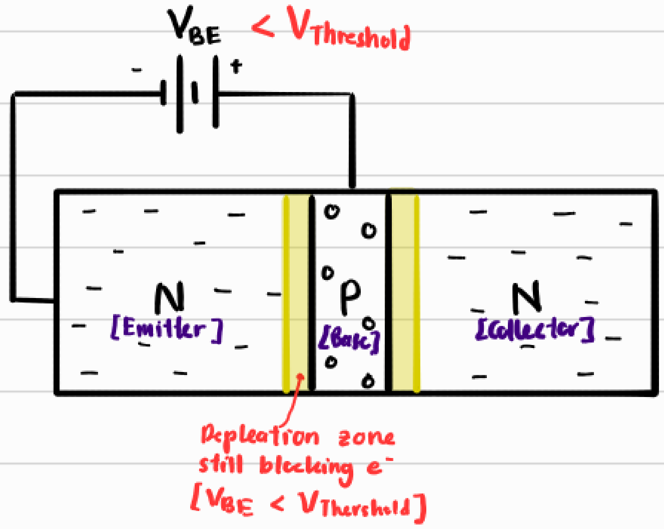
1) Essentially, when $V_{BE} > V_{Threshold}$, B-E depletion zone vanishes, e^- flows from Emitter to Base, thus supplying Base w e^-

2) Because B-C is in reversed bias, e^- in Base will get sucked into collector, right pass the depletion zone
[Just like Diode leak current]
$\Rightarrow I_c > 0$

3) Some e^- will recombine w the holes in Base, jumping from hole to hole, becoming base current.
$\Rightarrow I_B > 0$

① Cutoff [$V_{BE} < V_{Threshold}$]

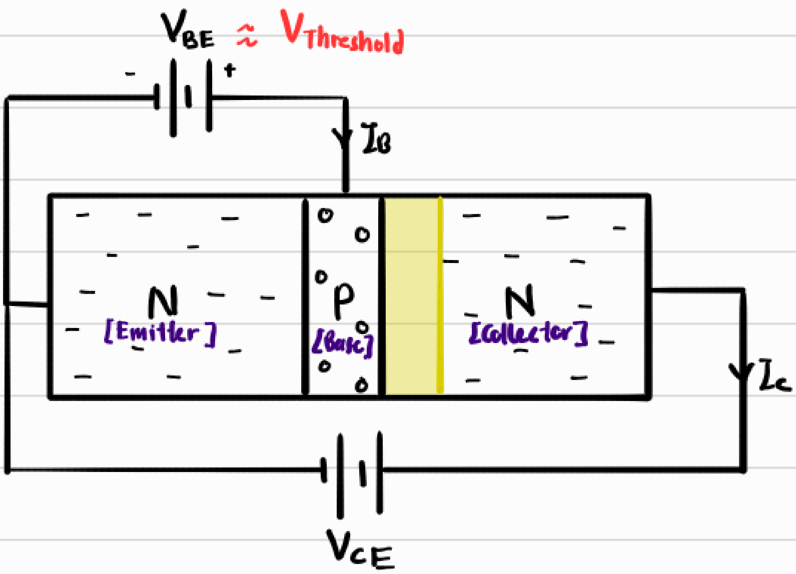
$V_{minimum}$ to overcome (depleation zone



∴ Since e^- can't go from Emitter to Base

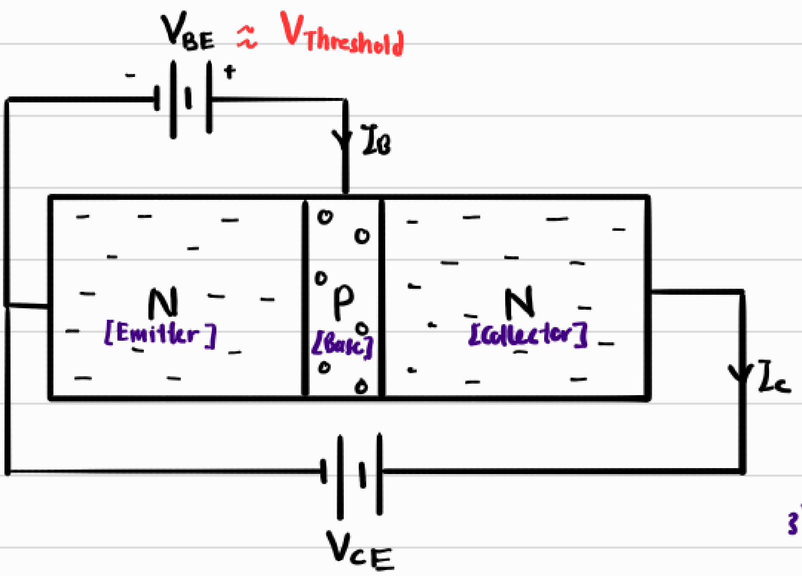
$\Rightarrow \boxed{I_c = 0}$

② Forward Active [$V_{BE} \approx V_{Threshold}$, $V_{CE} \geq V_{BE}$]



This is the perfect state where $\boxed{I_c = \beta I_B}$

③ Saturation [$V_{BE} \approx V_{Threshold}$, $V_{CE} \leq V_{BE}$]



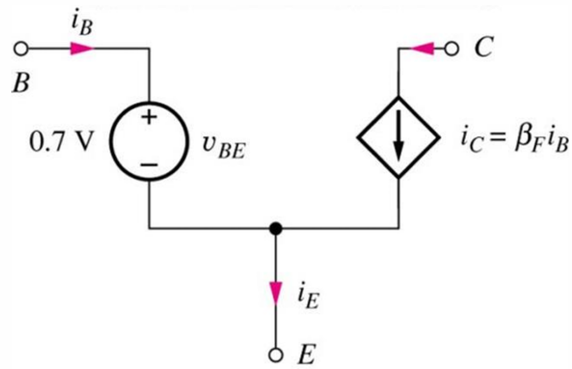
1) When $V_c < V_B$, B-C becomes forward bias

∴ e^- flow from C to B

2) Since e^- also flow from E to B, the Base will be saturated w e^-

3) In this state $\boxed{I_c < \beta I_B}$

Simplified Model for Forward Active (Current Controlled)



This is also known as "small-signal" model.

Base-emitter diode is replaced by constant voltage drop model (VBE = 0.7V) since it is forward-biased in forward-active region.

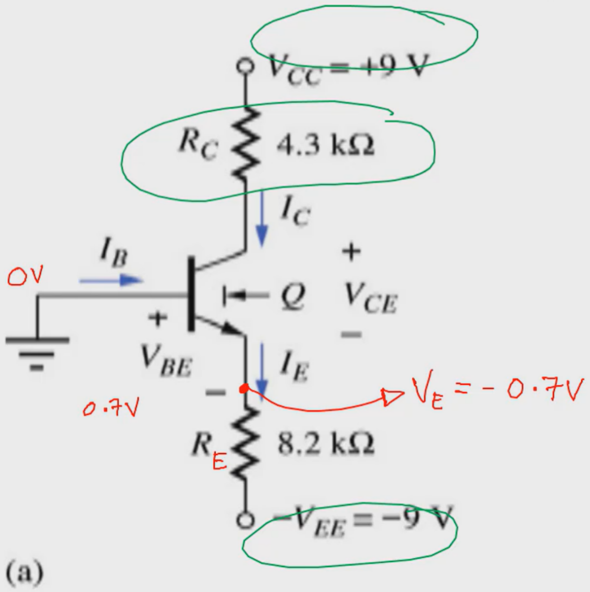
DC base and emitter voltages differ by 0.7 V diode voltage drop in forward-active region.

Collector Current equals forward current gain, βF, multiplied by base current.

Example 10.1

Find V_{CE} given that $\beta_F = 50$, $\beta_R = 1$

Assume: Forward-active region of operation, gives $V_{BE} = 0.7$ V



(a)

$$V_{RE} = -0.7 - (-9) = 8.3V$$

$$I_E = \frac{V_{RE}}{R_E} = \frac{8.3V}{8.2k\Omega} = 1.012 mA$$

$$\alpha_F = \frac{\beta_F}{\beta_F + 1} = \frac{50}{50+1} = 0.9804 = \frac{I_C}{I_E}$$

$$I_E = I_B + I_C = I_B + \beta I_B = (\beta + 1) I_B = (\beta+1)\frac{I_C}{\beta}$$

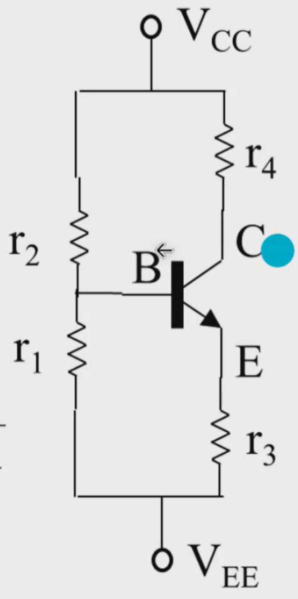
$$I_C = \alpha_F I_E = 9.922 \times 10^{-4} A$$

$$V_C = I_C \times R_C = 9.922 \times 10^{-4} \times 4.3k\Omega = 4.266 V$$

$$V_{CE} = 9V - V_C - V_E - (-9) = 5.434 V_{\#}$$

Four resistor biasing for BJTs

Start with an NPN transistor in a network of four resistors:

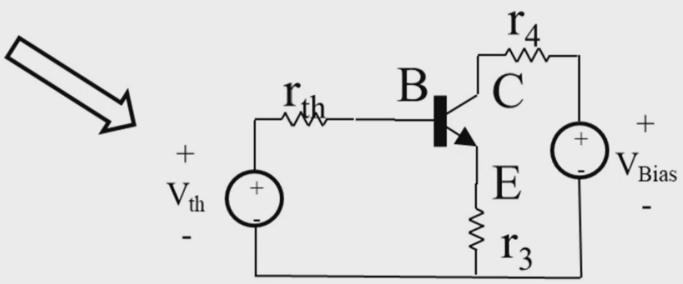


- Define voltage drop from V_{CC} to V_{EE} as bias voltage V_{bias}

$$V_{Bias} = (V_{CC} - V_{EE})$$

- Make the connection to base a Thevenin equivalent

$$r_{th} = r_1 || r_2 \quad V_{th} = (V_{Bias}) r_1 / (r_1 + r_2)$$

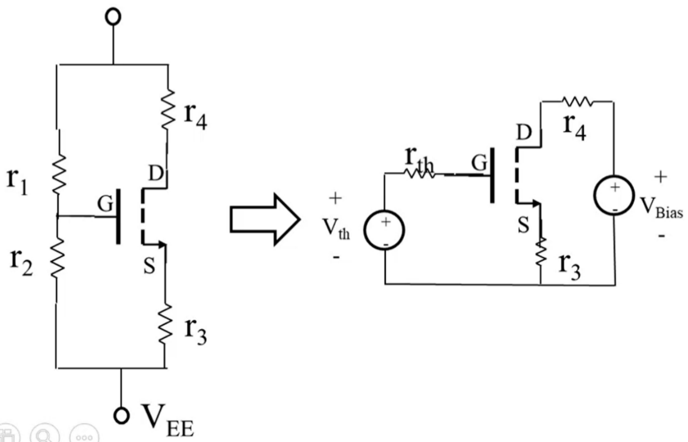


Biasing an NMOS transistor in saturation

For saturation region operation we require VDS ≥ (VGS – Vth)

To figure out if this is true, we start by assuming that the MOSFET is operating in this regime. If that assumption is false, then a different model of NMOS should be tried....

Start with an NMOS transistor in a similar network to before, simplify as before.



For MOSFETs, saturation implies that i_G=0, which then leaves i_S=i_D

i_D then has a set characteristic in saturation.